

3(a)	<i>HTT</i> / huntingtin ; A IT15 huntingtin ;	2
3(b)(i)	any one from: 1 (close) family history (of Huntington's) ; 2 adults that have (early) symptoms of Huntington's ;	1
3(b)(ii)	any two from: 1 adult-onset disease / symptoms generally occur later in life ; 2 adults can make lifestyle choices ; e.g. whether or not to have children 3 implications of positive result ; e.g. at-risk parent may not have taken test e.g. wait to see if parent has Huntington's 4 psychological impact / children not able to give (informed) consent / AW ; ora	2
3(c)	A DNA fragments or bands for alleles throughout 1 adult A has no alleles with more than 35 (CAG) repeats ; ora 2 adult B has one allele between 35 and 40 (CAG) repeats ; 3 adult C <u>and</u> adult D have one allele with more than 40 (CAG) repeats ; 4 adult D has more (CAG) repeats than adult C ;	4

Question	Answer	Marks
3(d)	any three from: 1 Huntington's disease is (autosomal) dominant ; 2 addition of normal allele only corrects recessive disorders or dominant allele cannot be, masked / switched off ; 3 gene therapy does not remove the, mutated / dominant, allele ; 4 not able to replace, nerve / brain cells ; 5 correct type of stem cells may not be available ; A correct ref. to viruses	3